



Arya Abhay Alvekar

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B.Tech - Electronics and Communication Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	9.03 (Current)	2022-Present
Senior Secondary	Maharashtra HSC Board	94.0%	2022
Secondary	ICSE Board	97.0%	2020

EXPERIENCE

- Google IT Services India Private Limited** May 2025 - Jul. 2025
Software Engineering Internship Bengaluru, India
 - Developed a **statistical** pipeline with a **tree-structured** architecture using **weighted averaging** and **selective data** processing on historical datasets, to enhance the accuracy of a consumer deliverable, achieving up to **10%** performance improvement.
 - Built workflows for **automated performance validation**, and a **dashboard** to visualize results across parameter settings.
- Distributed Co-Phasing for Federated Learning** Jul. 2025 - Dec. 2025
Bachelor Thesis Project under the guidance of Prof. Ribhu Chopra IIT Guwahati
 - Developed a system that leverages the **Distributed Co-Phasing** method of wireless communication for sensor data transmission and computing **over-the-air** global gradient used in **Federated learning**, with focus on practical applicability.

PROJECTS

- Reliable Data Transfer Protocols over UDP** Jan. 2025 - Feb. 2025
Course Project: EE 334 Computer Networks, IIT Guwahati [Github](#)
 - Independently implemented **Stop-and-Wait**, **Go-Back-N**, and **Selective Repeat** protocols from scratch over **UDP sockets**.
 - Developed a robust **client-server architecture** with custom packet structures, acknowledgments, and retransmissions.
 - Used **sliding-window** flow control to handle packet loss, ordering, and duplication, ensuring reliable data transmission.
- Decentralized Blockchain Simulator** Mar. 2025 - Apr. 2025
Course Project: EE 339 Blockchain Science and Technology, IIT Guwahati [Github](#)
 - Built a blockchain platform using **Python**, **Django** backend, and **ReactJS** frontend for decentralized operations.
 - Implemented **hashing**, **proof-of-work** mining, and transaction handling mechanisms to ensure secure block validation.
 - Achieved **multi-node consensus** with **RESTful APIs**, enabling synchronized blockchain state and real-time interactions.
- Embedded Systems Programming** Jul. 2024 - Dec. 2024
Course Project: EE 312 Embedded Systems, IIT Guwahati [Github](#)
 - Built a **real-time traffic light controller** in **ARMv7 assembly** on CPUlator, incorporating delay loops, LED sequencing, and countdown functionality on a **7-segment display**, showcasing **low-level** programming and hardware-software interaction.
 - Developed efficient **UART-based I/O routines** for **JTAG UART**, enabling dynamic text output with structured formatting.
- Other Projects**
 - Sign Language Interpreter** : Developed an **Arduino-UNO** based wearable using **flex sensors** to capture finger movements, enabling real-time sign prediction via **serial communication** using **Random Forest**, **Naive Bayes**, and **SVM** classifiers.
 - Weather App** : Built a website fetching real-time data using **OpenWeatherMap API**, featuring **HTML/CSS** interface.

TECHNICAL SKILLS

- Programming:** C/C++, Python*, Verilog*, GoLang*
 - Web Development:** HTML, CSS, JavaScript*, Django*
 - Data Analysis:** NumPy, Pandas, Matplotlib
 - Miscellaneous:** Git, $\text{\LaTeX}$, SQL*, Linux*
- * Elementary proficiency

KEY COURSES TAKEN

- Computer Science:** Data Structures & Algorithms, Computer Networks, Embedded Systems*, Blockchain Science & Technology*, Pattern Recognition & Machine Learning*, Hardware Aware Deep Learning
- Mathematics:** Probability & Random Processes, Linear Algebra & Differential Equations, Real & Complex Analysis
- Electrical and Electronics:** Principles of Communication*, Digital Circuits*, Digital Signal Processing, Applied Image and Signal Processing, Control Systems*, Analog Circuits, Semiconductor Devices, Electromagnetic Theory*

* Awarded AS/AA (10/10) Grade

ACHIEVEMENTS

- Joint Entrance Examination Advanced**, Secured **AIR 3216**, among **150,000+** candidates. 2022
- Joint Entrance Examination Main**, Secured **AIR 3396**, among **1 million** candidates. 2022
- Maharashtra State CET**, Secured **99.99** percentile, among **450,000+** candidates. 2022
- Inter-IIT Tech Meet 12.0**, Stood at **4th position** in the JLR Chiplet Challenge. 2023
- Meta Hacker Cup**, Secured rank **1700**, out of **13,000+** participants. 2025
- IICPC Codefest**, Secured prelims rank **1327**, out of **13,000+** participants. 2026

POSTION OF RESPONSIBILITY AND EXTRA CURRICULARS

- TA for Hardware Aware Deep Learning (DA O3021):** Aided in preparing question banks and curating course content.
- Classical Singing:** Completed 4 junior-level examinations with Akhil Bharatiya Gandharva Mahavidyalaya Mandal.
- Maharashtra State Elementary Drawing Examination:** Successfully cleared the examination.